

Assignment 3

Interactive Visualization with GitHub Pages Deployment

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Course	DS351 — Data Visualization

Dataset Description

The **Global Video Game Sales** dataset contains 16,000 records of video game titles released between 1980 and 2016 across 20+ gaming platforms. Each record includes the game name, platform, release year, genre, publisher, and regional sales figures (NA, EU, JP, Other) along with total global sales in millions of units.

Records	16,000 games
Columns	11 (Rank, Name, Platform, Year, Genre, Publisher, NA/EU/JP/Other/Global Sales)
Year Range	1980 – 2016
Genres	12 (Action, Sports, Shooter, Role-Playing, Platform, Racing, and more)
Platforms	20+ (PS2, X360, PS3, Wii, DS, NES, SNES, PC, and more)

GitHub Pages URL

https://muqeen-ahmed.github.io/DS351_Assignment_2/

Implemented Interaction Types

1. Select	Click a dot in the scatter chart to select and highlight that game across all views.
2. Filter	Genre checkboxes and dual year-range sliders filter the dataset across all charts simultaneously.
3. Encode	Dropdowns allow changing the X axis, Y axis, and color mapping (by Genre or Platform) of the scatter chart.
4. Connect	Selecting a point in the scatter chart dims unrelated bars and donut segments, linking all four charts.
5. Reconfigure	Buttons restructure the bar chart between three views: By Genre, By Platform, and By Year.

6. Elaborate	Clicking a dot opens a detail panel showing full game info and a regional sales breakdown.
7. Abstract	A toggle switches the scatter between individual games and genre-average aggregated points.

Charts

Bar Chart	Shows total sales grouped by Genre, Platform, or Year (switchable via Reconfigure buttons).
Scatter Plot	Plots individual games or genre averages across two user-selected regional sales axes.
Area Chart	Displays the global sales trend over time (1980–2016) with a gradient fill.
Donut Chart	Breaks down total sales by region: NA, EU, JP, and Other.